Army Report — Test Stand Delay Inputs

Nov 5, 2026 · @Mazen

Summary

The first Lake Clifton hot-fire is now scheduled for 29 Mar – 9 Apr 2027, after a cold-flow window of 9–25 Mar 2027. The hardware is not what held us back: the test stand is fabricated and tack-welded at McKinney Corp. The delay comes from the University's requirement that a licensed Professional Engineer certify every part of the facility before we build or test on University land.

That requirement turned into three separate engineering procurements, each run by MSU Procurement under Maryland procurement law:

1. **Structural PE** — seal the concrete foundation and anchorage (PWS-LC-FND-001, started July 2026).
2. **Fire protection PE** — independent hazard, quantity-distance and code review for LOX/RP-1 (PWS-LC-FPE-001, submitted 11 Sep 2026).
3. **Test stand PE** — review the steel stand design and sign off on its assembly (through McKinney, raised 30 Oct 2026).

Each one went through repeated revisions. Procurement changed the contracting method, asked for legal and liability language to be rewritten, and asked for University policy text to be quoted word for word. Site readiness added more time: demolition at Lake Clifton continues through 2026, and permanent electrical service is expected in early 2027.

NIAR: Dr. Busari to add NIAR's role and history here. Nothing on NIAR was found in the team's email or Drive.

Timeline

Since July 2026, most of the schedule has gone to University review and procurement cycles, not to engineering work. Newest first:

Date (2026)	Item	What happened
3 Nov	Fire protection PE	Procurement picks EBL Engineers and sends the scope out for an itemized quote. We are waiting for EBL to confirm receipt.
1 Nov	Test stand	Team member works at McKinney: the stand is cut, bolted, checked for dimensions and tack-welded.
30 Oct	Test stand PE	McKinney (Murph) offers a PE to review the stand design and sign off on assembly. The open question is whether the University accepts a PE licensed outside Maryland.

Date (2026)	Item	What happened
21 Oct	Fire protection PE	Dr. Busari sets the conditions for using EBL: LOX/RP-1 qualifications, our PWS as the contract scope, explosion/collapse/underground (XCU) insurance, and an itemized price.
20 Oct	Fire protection PE	Procurement switches from a full RFP to piggybacking on an Anne Arundel County Public Schools contract with EBL.
12–19 Oct	Fire protection PE	First RFP draft (27/PRO-0016-S) arrives with an old scope and IT-template text. The team returns corrections and Procurement reissues it.
7 Oct	Both PWSs	Procurement asks that all project risk sit with the vendor and that the vendor not interfere with other campus work. Both PWSs are revised again.
29 Sep	Structural PE	Technical review of the foundation design basis (Rev. 0A) finds 9 open items. Anchor changed to Hilti HDA.
11 Sep	Fire protection PE	Fire protection PWS Rev. 1 (47 pp) and Quantity-Distance Analysis (69 pp) submitted for University review. No reply until 7 Oct.
3–4 Sep	Structural PE	Procurement holds the structural PWS for legal and liability review. Rev. 2 adds an ownership statement, a responsibilities matrix and University policy text quoted word for word.
19 Aug	Site	Facilities approves moving the test area to Area 3. Demolition runs through 2026; permanent power comes in early 2027.
14 Aug	Structural PE	Procurement changes the method from Lowest Price Technically Acceptable (LPTA) to a Best Value RFP. The team must now serve on the evaluation committee.
1 Aug	Structural PE	Structural PWS Rev. 1 rewritten as a solicitation.
29 Jul	Structural PE	Foundation PWS (Rev. 0) sent to Procurement to start the search for a PE.
1 Jul	Test stand	Waterjet quote of $10,734.69 from Metal Supermarkets, which could not do the corner welds. A full-service fabricator was needed.

NIAR

For Dr. Busari to fill in. I found no NIAR emails, files or project notes in my records, so this section needs your input.

- ☐ What NIAR was contracted or asked to do, and when
- ☐ What was delivered or not delivered, and how that affected the schedule
- ☐ Current status and whether NIAR is still involved

McKinney Corp

McKinney Corp is fabricating the steel test stand, and the stand is now physically built and tack-welded. McKinney was chosen because our first quote (Metal Supermarkets, 1 Jul 2026, $10,734.69 for waterjet-cut plates) could not cover the corner welds. We needed a shop that could cut, weld, assemble and finish in one place.

Status as of the 1 Nov 2026 shop visit:

- Tubes cut, structure bolted together following about 95% of our assembly sequence, then checked for squareness, parallelism and flatness before tack welding. The technician will finish the final welding.
- McKinney can also powder-coat the stand, roll the fin can and covers, make carbon-fiber tubes and the nose cone, machine the manifolds, and clean parts ultrasonically for LOX service.
- One issue was recorded. Parts we had put on hold for design changes were waterjet-cut anyway. McKinney asked us to send the final changes, and the revisions are going ahead.

How McKinney can help with certification: on 30 Oct 2026 we proposed that McKinney assemble the stand in their shop, ship it, and reassemble it on site. Murph (McKinney) offered to find a PE who will review the full stand design against our expected loads, sign off on the assembly, and take responsibility for the whole stand. A cost estimate is coming. The open question is whether the University will accept a PE licensed outside Maryland. The foundation PWS requires a Maryland license at the time of sealing, which can be obtained through comity (a state accepting another state's license).

Structural PE certification

The concrete foundation and anchorage can't be poured until a Maryland-licensed structural PE verifies, seals and certifies the design. The team's engineering is finished. What remains is hiring and contracting that PE, and that process has taken four months so far.

What the PE certifies: a reinforced-concrete pad with 16 Hilti HDA-TR M12 stainless undercut anchors. In our analysis, overturning governs at a factor of safety of 3.52 (1.76 with a dynamic amplification factor of 2.0). The PE also covers drainage and spill containment: a level deck, a sloped apron, a trench drain and a buried double-wall tank of about 300 gal.

Why the procurement took this long:

Revision	Date (2026)	Reason for the change
Rev. 3	Nov	Combines the solicitation framework, technical basis and drainage. Adds vendor risk and non-interference clauses (Procurement, 7 Oct) and itemized pricing (3 Nov).
Rev. 0A	29 Sep	Technical review found 9 open items. A drop-in anchor rated for non-structural use only was replaced.
Rev. 2	4–6 Sep	Procurement's legal and liability review: ownership, a responsibilities matrix, and University insurance and indemnity text quoted word for word.
Rev. 1	1 Aug	Rewritten as a solicitation. The method then changed from LPTA to a Best Value RFP on 14 Aug.
Rev. 0	29 Jul	Design basis sent to Procurement to start the PE search.

Still open with Procurement: the procurement method, the errors-and-omissions insurance limit, the solicitation numbers, the issue date and the exact General Conditions text.

Fire protection PE certification

No propellant can be stored or fired on site until a Maryland-licensed fire protection engineer independently reviews the design and issues signed and sealed documents. The Baltimore City Fire Marshal must also approve the site. As of 3 Nov 2026, Procurement has picked EBL Engineers and asked for an itemized quote. EBL had not yet confirmed receipt.

What the engineer reviews:

- 38.5 lbm RP-1 and 99.4 lbm LOX on the stand for each firing; ready storage of 1,157 lbm LOX and 370 lbm RP-1.
- Firings: about 5 s to verify ignition and 12.4 s to depletion, priced on a 30 s envelope.
- Applicable codes: the IFC 2021 hazardous-materials chapters, NFPA 55 and 30, quantity-distance siting, and NFPA 15 water spray.
- Eight deliverables (D1–D8): site assessment, code and hazard analysis, fire protection design, emergency plan, agency coordination, and sealed documents with acceptance records.

Why it took this long: the team submitted a complete package on 11 Sep 2026 (a 47-page PWS and a 69-page Quantity-Distance Analysis) and got no response for four weeks. The first RFP draft came back on 12 Oct with the old scope values and text copied from an IT/software template, such as an "Enterprise Ticketing Platform" and a SOC 2 audit. On 20 Oct, Procurement dropped the RFP and chose to piggyback on an existing school-district contract with EBL. EBL's experience is mainly in school buildings, so Dr. Busari made the award conditional on EBL meeting the PWS qualifications, or adding a specialty subconsultant who does.

Certification of every part of the project

Three of the seven parts below need an outside licensed engineer, and none of those three is under contract yet. That is the critical path to hot-fire.

Part	Who certifies	Status (5 Nov 2026)
Concrete foundation and anchorage	Maryland-licensed structural PE	Design and PWS Rev. 3 complete. Waiting on Procurement to issue the solicitation.
Fire protection, siting and propellant storage	Maryland-licensed fire protection PE (EBL Engineers)	Itemized quote requested 3 Nov. Waiting for EBL to confirm.
Steel test stand design and assembly	PE arranged through McKinney	Fabricated and tack-welded. Waiting on the PE cost estimate and a University decision on an out-of-state PE.
Site layout (Area 3)	MSU Facilities	Approved 19 Aug with a 20 ft offset along the north curb.
Site electrical service	MSU Facilities	Expected early 2027. Until then, an 11 kW generator.
Propellant handling procedures (SOPs)	Team, then the fire protection PE	In progress. SOP 03 covers LOX receiving, dewar handling, loading, vent and dump.
Vehicle fill and quick-disconnect interface	Team	LOX/RP-1 flange-clip QDs bench-tested at 80 psi with clean self-ejection.

Email excerpts

These short quotes, taken word for word from the project threads, show where time was spent. Dr. Busari can choose which ones to include.

Procurement, 14 Aug 2026 (structural PWS):

> *"After further review of the PWS, given the scope's complexity, Procurement will be using an RFP with a Best Value evaluation approach rather than LPTA."*

Procurement, 3 Sep 2026 (structural PWS):

> *"I believe the document warrants further administrative and legal review before it is finalized. The concern is that the current language appears to place a significant level of responsibility and potential liability on Morgan State University..."*

Dr. Busari to Procurement, 14 Sep 2026:

> *"The president is going to request updates on our testing schedule come this fall and we have not broken ground yet."*

Team to Procurement, 7 Oct 2026 (fire protection):

> *"On September 11, Noureldin submitted the Fire Protection Performance Work Statement... As we have not yet received an update, we wanted to check on the current status."*

Team review of the first fire protection RFP, 14 Oct 2026:

> *"Several parts appear to come from an IT/software RFP template: the 'Enterprise Ticketing Platform' in Section 1.1, the SOC 2 Type 2 audit in Section 5.2..."*

Procurement, 20 Oct 2026:

> *"Instead of initiating a multi-month RFP cycle, we are actively evaluating 'piggybacking' off the existing Anne Arundel County Public Schools contract with EBL Engineers, LLC."*

Dr. Busari, 21 Oct 2026:

> *"EBL's fire protection experience appears to be in school buildings. Our job needs LOX/RP-1 hazard work..."*

McKinney meeting summary, 30 Oct 2026:

> *"Murph contacted Yusuf and told him that he has a PE who can sign off on the project. He will estimate the cost and get back to Yusuf."*

Facilities, 19 Aug 2026 (site):

> *"Demolition will be ongoing throughout this year and electric service will likely be installed early next year."*

Figures and images

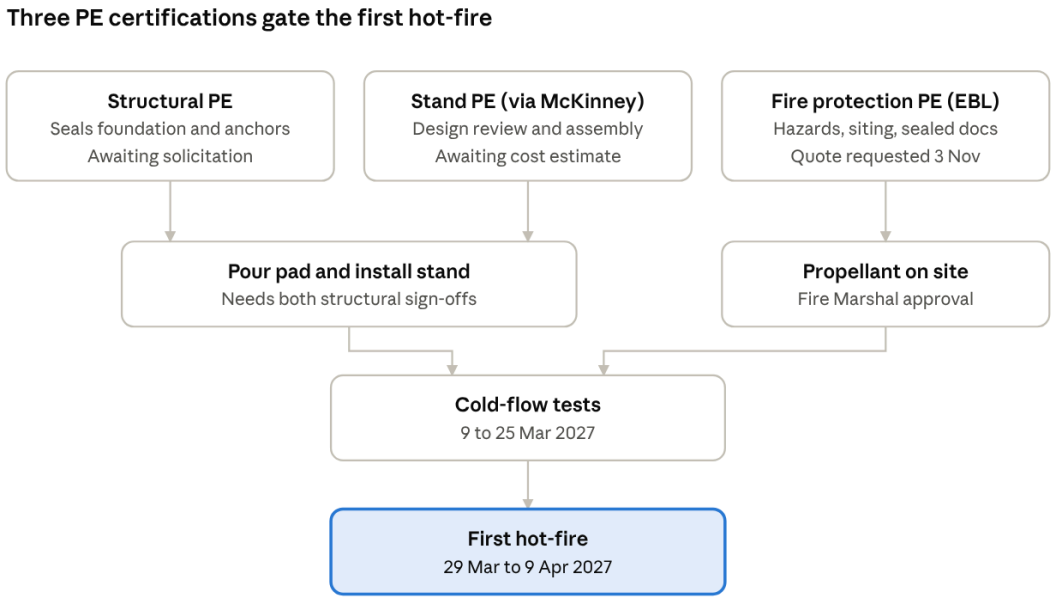
The strongest visual evidence is the McKinney shop photos from 30 Oct – 1 Nov 2026. They show the stand assembled on its inspection supports and tack-welded. The table below lists what is ready to pull into the report.

Figure	What it shows	Where
McKinney shop photos and videos (about 30 files)	Stand cut, bolted, inspected and tack-welded; rolled shell; composite parts	Drive folder
Test pad with buried tank	Foundation, sloped apron, trench drain, ~300 gal containment tank	Aero project: Test_Pad_Current_Design_with_Buried_Tank
Test stand 3D model	Steel stand geometry	Aero project: Rocket_Test_Stand_3D_Model

Figure	What it shows	Where
Foundation anchor design package	16-anchor Hilti HDA layout and checks	Aero project: Foundation_Anchor_Design_Package
Test pad engineering plan	Full pad design basis	Aero project: Rocket_Test_Pad_Engineering_Plan.pdf
Structural PWS Rev. 3	The document waiting for a PE	PDF
Site meeting slides, 18 Aug 2026	Area 3 layout, site constraints	Attachment to Noureldin's 18 Aug email to Angelo Pirali
Site power plan and panel locations	Planned permanent power for early 2027	Attachments to Angelo Pirali's 19 Aug email
Fire protection PWS Rev. 1 and QD Analysis	47 pp scope and 69 pp siting analysis	Attachments to Noureldin's 11 Sep email

Path to first hot-fire

The March–April 2027 windows hold only if all three PE contracts are signed soon. Each review, the pad construction and the Fire Marshal approval all have to finish before cold-flow begins on 9 Mar 2027.



path to first hot-fire · 3 certifications, 2 site gates

Construction can't start without the structural sign-offs, and no propellant can come on site without the fire protection engineer's sealed documents.

Actions that would shorten the path:

- ☐ Procurement issues the structural PWS Rev. 3 solicitation.
- ☐ EBL submits its itemized quote, shows LOX/RP-1 qualifications (or adds a specialty subconsultant), and a contract is signed.
- ☐ The University decides whether it accepts an out-of-state PE for the stand. The team sends the stand loads to McKinney.
- ☐ Facilities confirms the date for permanent site power.